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Architecture Selected:

Layered Architecture for the Food Surplus Redistribution Platform.

Reasons:

Layered Architecture is suitable because the system can be divided into separate logical layers based on their responsibilities. The User Interface handles interactions with Donor Partners and Shelter Coordinators, while components such as Food Management, Reservation, and Notification & Pickup handle the core business operations. The Food Bank Database is responsible for storing food listings, expiry information, reservations, and user-related data. The first reason for selecting this architecture is clear separation of responsibilities. Food listing and expiry tracking can be handled by the Food Management component, while reservation and pickup-related operations can be handled separately. This makes the system easier to understand, develop, test, and maintain. The second reason is easier management of the food redistribution workflow. Requests from users can pass through the appropriate business components before accessing the database. For example, a shelter can request available food through the User Interface, reserve a food batch through the Reservation component, and complete pickup verification through the Notification & Pickup component.

Security Advantage:

Layered Architecture provides a security advantage by preventing the User Interface from directly accessing the database. Database operations can be controlled through the business components, allowing the system to validate requests and restrict unauthorized access to food listings, reservations, and user information.

Performance Benefits:

Layered Architecture can improve performance by separating frequently used operations into appropriate components. Food listing, reservation, notification, and pickup operations can be processed independently within their respective components, reducing unnecessary processing and allowing requests to be handled efficiently.

Conclusion:

Layered Architecture provides a clear, organized, and maintainable structure for the Food Bank Surplus Redistribution Platform while supporting security and efficient processing of its core operations.

Food Bank Surplus Redistribution Platform

